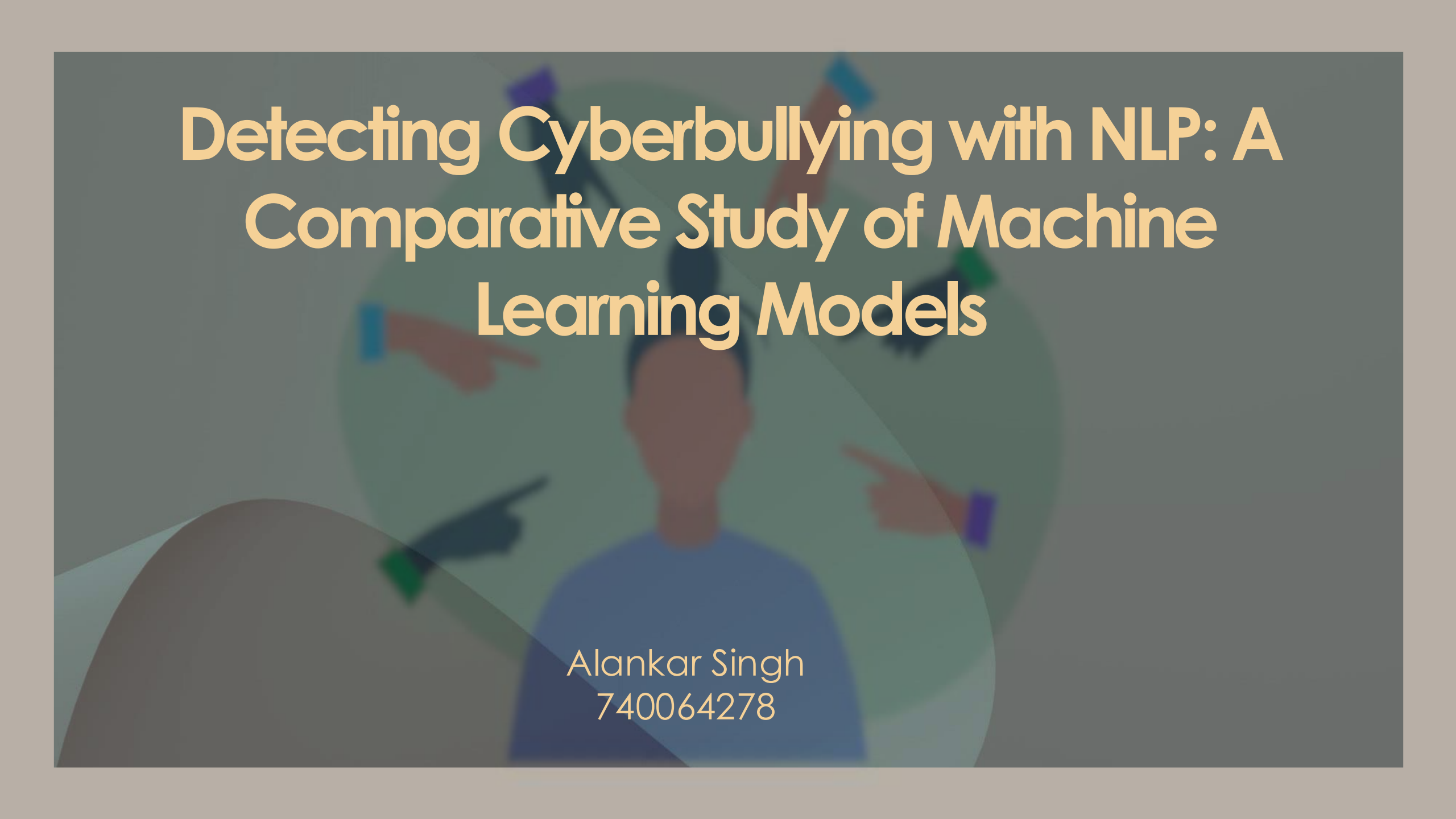


Detecting Cyberbullying with NLP: A Comparative Study of Machine Learning Models

The background features a stylized illustration of a person with dark hair and a blue shirt, positioned centrally. Surrounding this person are several hands of different colors (red, blue, green, purple) pointing towards them, symbolizing cyberbullying. The entire scene is set against a dark, muted green background with a subtle circular pattern.

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Why do we need to tackle cyberbullying?

Global Prevalence:

- According to studies, more than 37% of youth have been victims of cyberbullying, and this number is rising.
- 60% of internet users have witnessed some kind of online harassment, according to a Pew Research Centre study.

Psychological Impact:

- Suicidal thoughts, anxiety, and depression
- Damage to mental health, self-esteem, and social interactions

Ineffectiveness of manual moderation:

- Social media organizations employ human moderators, but the process is costly, time-consuming, and prone to bias.

Need for an automated system

- Given the growing number and rate of cyberbullying episodes, machine learning (ML) and natural language processing (NLP) tools offer an effective approach to identify and classify harmful online behaviour.
- AI-powered models can quickly analyse millions of messages in real time.
- Machine learning models can monitor vast amounts of social media content without human intervention.
- An automated system can flag harmful content before it causes damage to users.

Our Dataset

Dataset Overview

Total
Entries

47,692
tweets

Missing
values

None

Mean text
length

136
characters

Max text
length

5,018
characters

Columns

tweet_text
(the actual tweet)

cyberbullying_type
(classification label)

Class labels

Religion

Age

Gender

Ethnicity

Other cyberbullying

Not cyberbullying

Class balance

Cyberbullying type	Number of tweets	Percentage
Religion	7,998	16.7%
Age	7,992	16.7%
Gender	7,973	16.7%
Ethnicity	7,961	16.6%
Not cyberbullying	7,945	16.6%
Other cyberbullying	7,823	16.4%

Proposed approach

Text Preprocessing and Feature Extraction

Text preprocessing

Tokenisation

- Splitting text into individual words or subwords.
- Helps break tweets down into meaningful components.

Lowercasing

- Converting all text to lowercase.
- Ensures words such as “HATE” and “hate” are treated the same.

Text preprocessing

Stopword removal

- Removing common words (“the”, “and”, “is”) that don’t carry meaning.
- Helps in reducing dimensionality and improving efficiency.

Punctuation and special character removal

- Eliminating symbols (@, #, !, ?) that don’t add value.
- Helps in preventing unnecessary noise in text.

Text preprocessing

Stemming

- Reducing words to their root form ("running" -> "run") using Porter Stemmer.
- Helps normalise texts while preserving meaning.

Lemmatization

- Converting words to their dictionary form ("better" -> "good", "went" -> "go").
- Helps improve understanding of words in context.

Feature Extraction

Bag of words

- Creates a matrix counting word occurrences in tweets.
- Captures common bullying-related words like “idiot”, “loser”, “attack”.

TF-IDF (Term Frequency-Inverse Document Frequency)

- Assigns importance to words by reducing influence of common terms.
- Helps identify key bullying terms without biasing toward frequent words.

Feature Extraction

Parts of Speech (POS) Tagging

- Identifies adjectives, verbs, nouns, and pronouns.
- Aggressive tweets often contain specific POS pattern (e.g., insults)

Chunking

- Groups words into phrases (e.g., “stupid idiot” -> insult phrase).
- Helps detect aggressive language patterns.

Feature Extraction

VADER Sentiment score

- Classifies tweet as positive, neutral, or negative.
- Cyberbullying tweets are often negative or aggressive.

TextBlob Sentiment Score

- Provides polarity score (-1 = negative, +1 = positive)
- Helps detect intent behind a tweet.

Feature Extraction

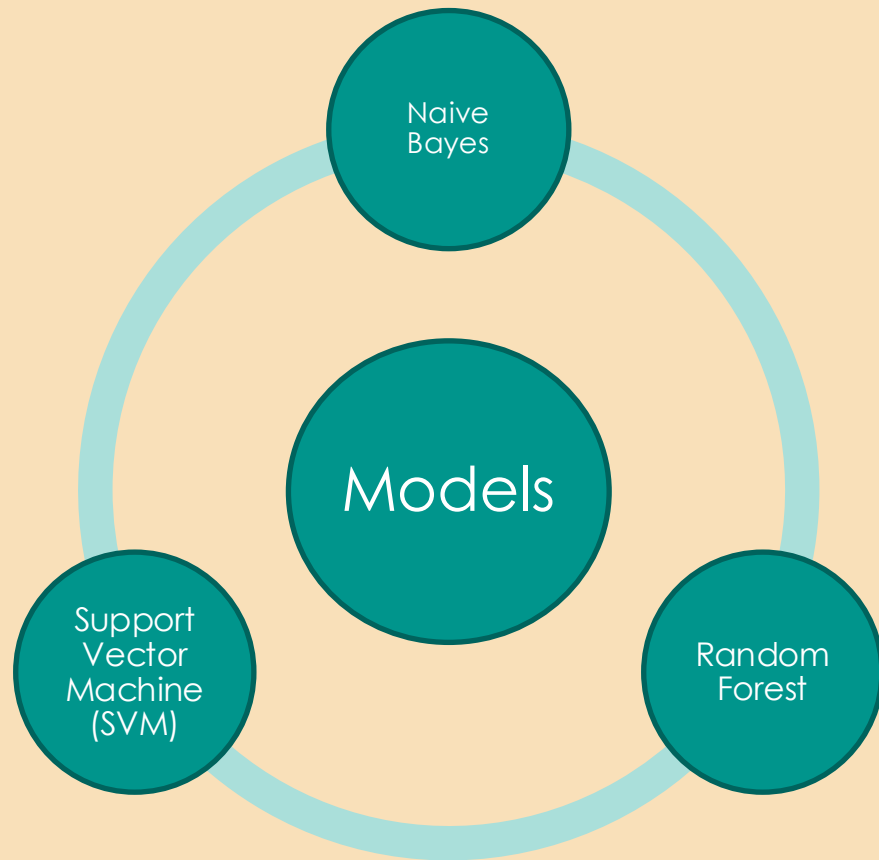
Named Entity Recognition (NER)

- Identifies people, locations, organizations, religions, and nationalities.
- Detects targeted cyberbullying cases involving specific individuals or groups.



Machine Learning

Models and the key parameters used



- Naive Bayes: Works well with Bag-of-Words (BoW) and TF-IDF for text-based classification.
- Support Vector Machine (SVM): Effective in high-dimensional text spaces.
- Random Forest: Captures complex patterns in cyberbullying language.

Limitations

Data limitations:

- Presence of very short & long tweets
- Slang, misspellings, and informal language
- Hashtags & mentions may interfere

Possible solutions:

- Remove outliers (e.g., tweets with <5 characters).
- Use lemmatization & slang dictionaries.
- Remove hashtags or analyse separately.

Thank You

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